Weekly Meeting

25.07.11

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Paper review

- Stureborg et al.,(2024)
  - https://arxiv.org/pdf/2405.01724

- Used the SummEval dataset w. human-annotated summaries.
  - *only single scoring w.o. pair-wise
- Evaluated using GPT-3 and GPT-4 as evaluators.
- For each summary:
  - Evaluation score (on a 1–5 scale).
  - Measured the perplexity.

Paper review

- Result

Score	Mean perplexity
1	8.42
2	7.53
3	7.33
4	6.44
5	5.51

- Lower perplexity → Higher evaluation score
- Reflects a preference for familiar or more predictable text

Paper review

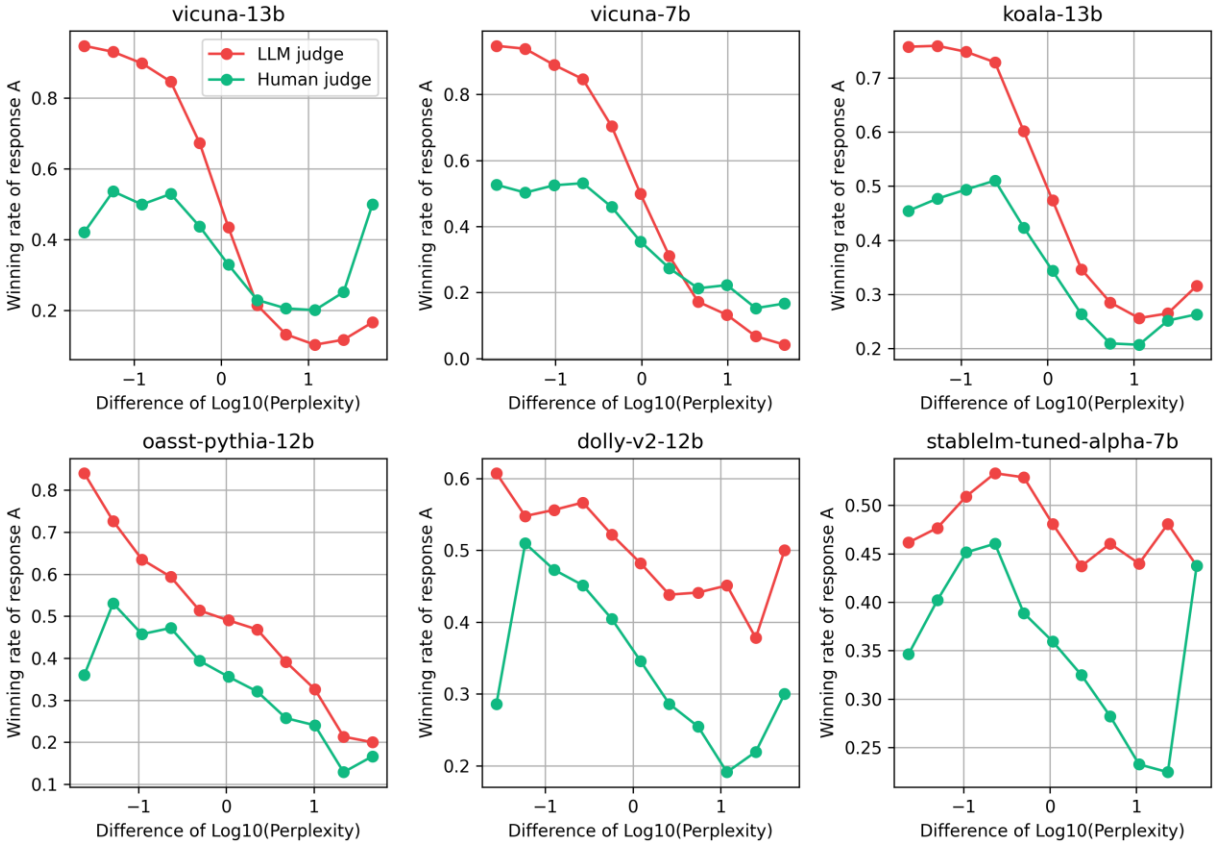
- Wataoka et al. (2024)
  - https://arxiv.org/html/2410.21819v1

- Used the Chatbot Arena dataset w. pair-wise

- Evaluated GPT-3.5 turbo, GPT-4 and others.
- For each summary:
  - Pair-wise evaluation
  - Measured the perplexity.

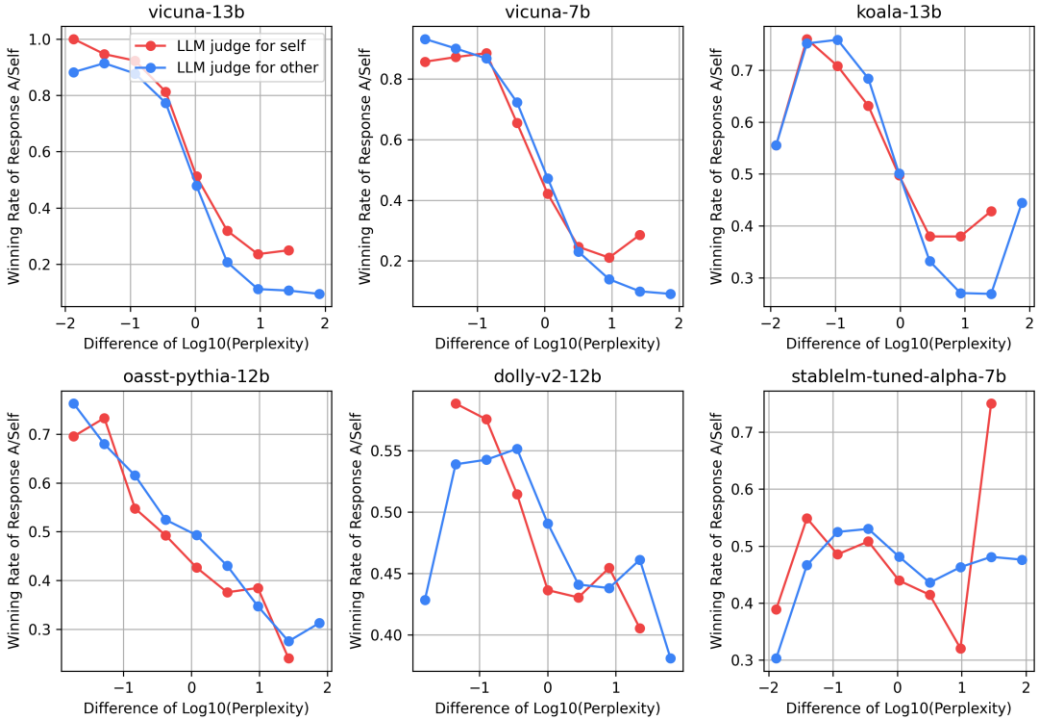
Paper review

- Result
- Correlation of perplexity & pair-wise win rate



Paper review

- Result
- LLM evaluators assign higher ratings to texts with lower perplexity
- regardless of whether the text was self-generated or produced by other models.



Experiment setting

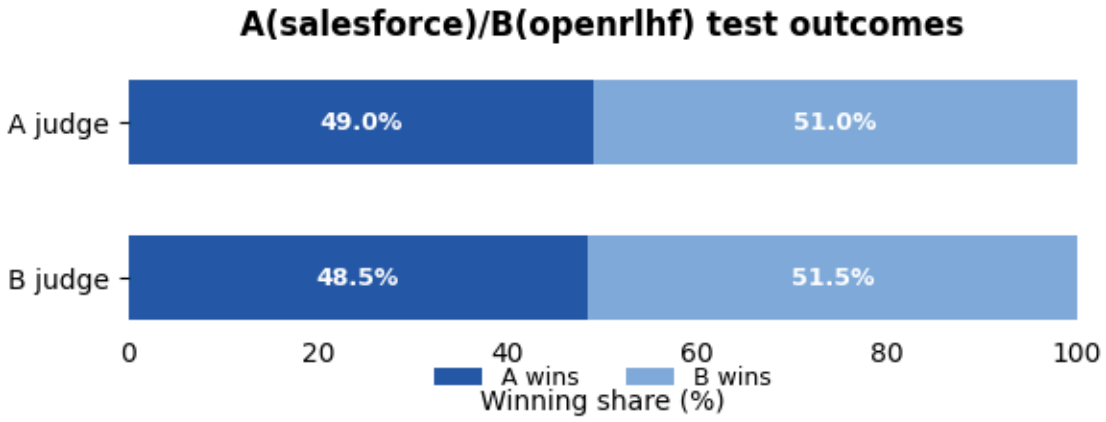
- Benchmark: AlpacaEval

- Only pair-wise test(A/B test)

- Models:
  - Llama3
  - Mistral
  - Qwen2.5

Result

- Llama3
  - Salesforce/LLaMA-3-8B-SFR-Iterative-DPO-R (A)
  - OpenRLHF/Llama-3-8b-iter-dpo-179k (B)

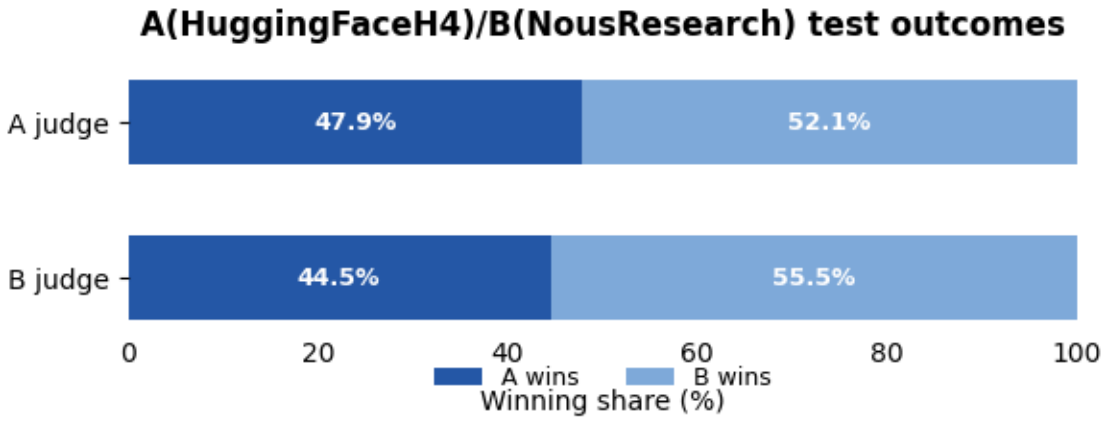


	Number
Bias	26
non-bias	23

*Bias: Judge prefer itself
non-bias: Judge prefer each other

Result

- Mistral
  - HuggingFaceH4/zephyr-7b-beta (A)
  - NousResearch/Nous-Hermes-2-Mistral-7B-DPO (B)

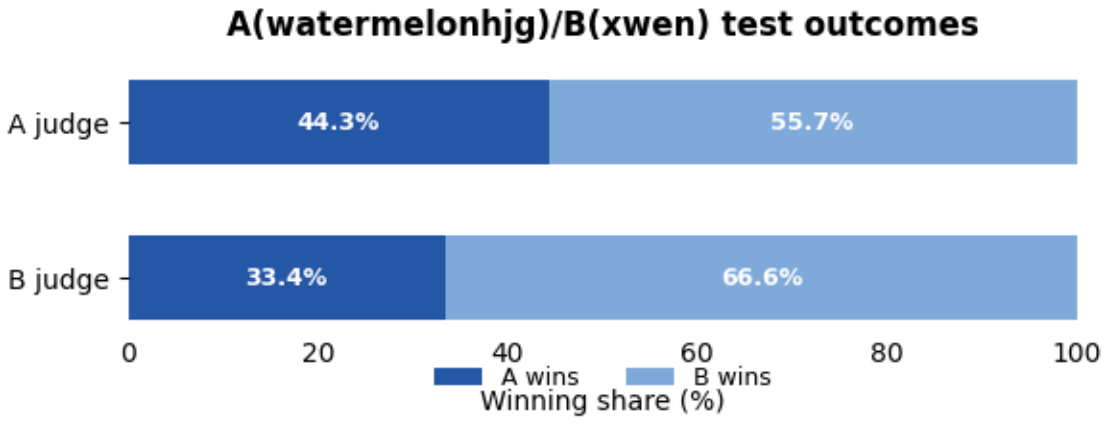


	Number
Bias	112
non-bias	86

*Bias: Judge prefer itself
non-bias: Judge prefer each other

Result

- Qwen2.5
  - watermelonhjg/Qwen2.5-7B-EN-Zero (A)
  - xwen-team/Xwen-7B-Chat (B)

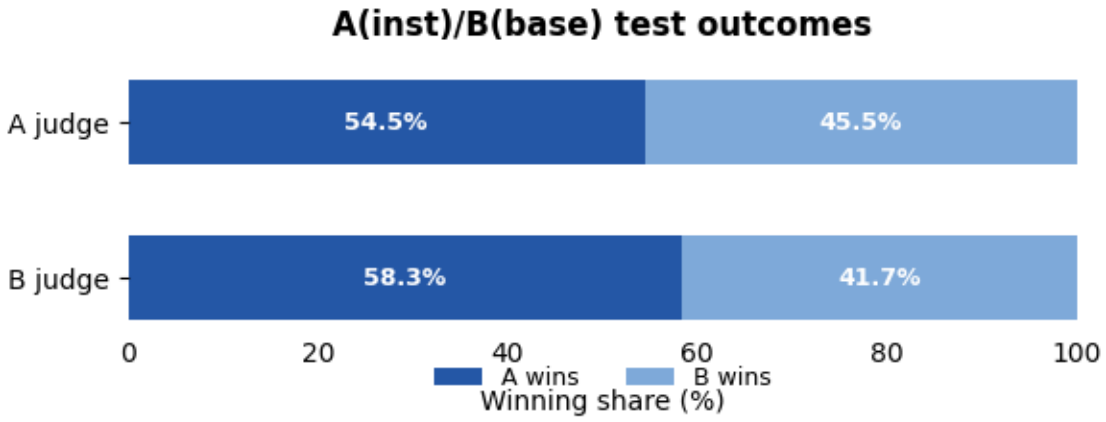


	Number
Bias	68
non-bias	27

*Bias: Judge prefer itself
non-bias: Judge prefer each other

Result

- Llama3 (DPO base)
  - Meta-Llama-3-8B-Instruct (A)
  - Meta-Llama-3-8B (B)



	Number
Bias	95
non-bias	88

*Bias: Judge prefer itself
non-bias: Judge prefer each other

Discussion

- Different method of tuning
  - 데이터/학습 방법에 따른 정도의 차이 존재
  - Bias의 경향성 확인

- DPO base
  - Base/inst 모델의 확률 차이에 따른 reward 계산
    - Pair-wise test에서의 선호 차이와의 상관관계

$$r(x, y) = \beta \log \frac{\pi_r(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x).$$